

Gourav Dangi

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EDUCATION

Indian Institute of Technology, Guwahati
B.Tech in Civil Engineering

2019 - 2023
CGPA: 7.34

EXPERIENCE

- Respo Financial Limited** Jun. 2025 - present
AI ML Engineer website
 - Built a **hierarchical RAG-based customer support automation system** that classifies inbound tickets and composes grounded first responses, automating first-response handling for **77% of total support tickets**.
 - Reduced first-response turnaround time from **48 hours to under 1 minute** while generating contextual **response notes and CX summaries** for unresolved tickets to reduce manual investigation effort.
 - Designed retrieval, LLM orchestration and **category-specific workflow automation** across support systems, integrating local LLM inference via **Ollama** for cost-efficient on-prem model serving.
 - Developed **AI voice-calling agents** using **Sarvam STT/TTS** for debt collection, EMI reminders and customer assistance, completing **3,000+ automated calls daily equivalent to the workload of 30+ human agents**.
- JM Financial** Jun. 2023 - May. 2025
Data Scientist website
 - Created and deployed **RAG chat-bot** using spacy for **intent classification and entity recognition** that resolve preliminary customer-support queries, that **resolves 86% of user queries that redirects to chat-agent**.
 - Implemented **document vectorization** and namespace identification, enabling the system to answer queries based on **financial documents and compliance reports** while delivering actionable insights.
 - Designed and implemented a **Graph Neural Network** model to effectively capture complex relationships within a user-product interaction network. Built a graph representation to analyze connections and **predict link existence**.
 - Developed a **Lead Scoring model** with **high recall** using advanced statistics and machine learning, boosting **conversion rates by 10%** through targeted marketing and **optimized lead prioritization**.

PROJECTS

- Self-Learning AI Agent with Reinforcement** Jul. 2026 - Aug. 2026
Personal Github
 - Built a local-first **self-learning agent** that improves across sessions without fine-tuning via **retrieve → act → evaluate → reflect → write**, using **Mem0, Qdrant, and Ollama**.
 - Designed **typed memory (facts, lessons, skills)** so user facts, feedback-driven behavior updates, and promoted skill playbooks retrieve independently instead of mixing in one vector dump.
 - Implemented **hybrid retrieval (vector score + lexical overlap + query expansion + domain gating)** and a **fact-correction** path that deletes contradicted memories and writes replacements from user notes (human feedback).
- Multi-Armed Bandit for Nudge Time Prediction** Mar. 2024 - Aug. 2024
Personal
 - Developed a solution to **identify best time to trigger nudge** so that user is most likely to interact with it.
 - Using **Reinforcement Learning** from Human Feedback technique **increased CTR from 2.4% to 7.4%**.
- Anomaly detection using Speech-to-Text and conversation summarization** Dec. 2023 - Mar 2024
 - Analyzed call recordings** between sales agents and customers using **speech-to-text** models such as **Whisper**.
 - Developed an **automated pipeline** leveraging **call transcripts** to detect potential mis-selling through **LLM**.
 - Generated detailed **reports and dashboards** to assist sales manager in identifying and addressing these mis-selling.

TECHNICAL SKILLS

- Programming:** Python, Typescript, SQL, C++
- Database:** PostgreSQL, MySQL, MongoDB, Pinecone, Neo4j
- AI / ML:** PyTorch, TensorFlow, scikit-learn, spaCy, LangGraph, SmolAgents, NLP, RLHF
- GenAI & AI Systems:** RAG, Fine Tuning, AI Agents, Ollama, Prompt Engineering, STT/TTS Systems, Qdrant
- Core Competencies:** Applied AI, Conversational AI, Voice AI, Latency Optimization, Graph ML, Model Deployment

ACHIEVEMENTS

- BlinkX Rodies Achiever Award** equivalent to **employee of the quarter** in respective domain. Q4 - 2023
- Institute Secretary** elected to be secretary of volleyball club for 2020-21. 2020
- Competitive programming platform** rated **1800+** on Leetcode. Leetcode
- Inter-IIT Cultural meet** secured **3rd position** among 23 IITs in Theater Module held at IIT Bombay. 2019